

## **Review Article**

# **Inositol Polyphosphate 5-Phosphatases: Lipid Phosphatases With Flair**

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### **Summary**

Recent studies have identified the inositol polyphosphate 5-phosphatases as a large family of signal modifying enzymes comprising 10 mammalian and 4 yeast family members. A number of investigations including gene-targeted deletion of 5-phosphatases in mice have demonstrated that these enzymes regulate many important cellular events including hematopoietic cell proliferation and activation, insulin signaling, endocytosis, and actin polymerization.

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**Keywords** Actin cytoskeleton; B cell proliferation; calcium signaling; inositol 5-phosphatase; insulin signaling; vesicle trafficking.

### **CELLULAR FUNCTIONS OF INOSITOL POLYPHOSPHATE 5-PHOSPHATASES**

The phosphatidylinositols are phospholipids present in all mammalian cells, both on the inner wall of the plasma membrane and in various discrete subcellular compartments such as the Golgi, endoplasmic reticulum, endosomes, and nuclear membrane. In response to agonist stimulation by growth factors, neurotransmitters, hormones, and peptide agonists, phosphatidylinositols are phosphorylated by specific kinases and thereby generate multiple intracellular signals that include <sup>1</sup>PtdIns(4,5)P<sub>2</sub>, PtdIns(3,4,5)P<sub>3</sub>, and PtdIns(3,5)P<sub>2</sub>. These signaling molecules induce a vast number of significant cellular events mediated by recruiting various effector proteins to the membrane and thereby contributing to their activation (Fig. 1;

1–7). PtdIns(4,5)P<sub>2</sub> regulates actin polymerization, vesicular recycling, and serves as a precursor for Ins(1,4,5)P<sub>3</sub> mediated-intracellular calcium release. PtdIns(3,4,5)P<sub>3</sub> regulates cell proliferation, cell death, differentiation, actin polymerization, vesicular trafficking, insulin signaling, cell migration, and secretion. The inositol polyphosphate 5-phosphatases (5-phosphatases) dephosphorylate 5-position phosphoinositide and inositol phosphate messenger molecules resulting in either the generation of potentially new signaling molecules such as PtdIns(3,4)P<sub>2</sub>, PtdIns 4-P, and PtdIns 3-P (Figs. 1 and 2), or molecules such as Ins(1,4)P<sub>2</sub> or Ins(1,3,4)P<sub>3</sub>, which no longer mobilize intracellular calcium but may lead to other signaling pathways (Fig. 3). Therefore, the 5-phosphatases may act as signal generating, or signal terminating enzymes (8, 9).

The investigation of 5-phosphatase enzyme cellular function is complicated by the vast number of agonists that generate phosphoinositide synthesis; the diverse roles these various phosphoinositide signaling molecules play in cellular activation; the significant number of 5-phosphatases that have recently been identified, many of which are expressed in the same cell; the distinct subcellular localization of the various isoforms; the diverse number of potential substrates; and the difficulties in discriminating between the results of 5-phosphatase substrate specificity shown by in vitro enzyme assays using recombinant 5-phosphatases, versus the results shown by in vivo cellular models in which 5-phosphatases have been either overexpressed or underexpressed by gene-targeted deletion in mice, or by anti-sense studies in cell lines. This review focuses on recent studies that reveal the distinct functional roles specific 5-phosphatases play in regulating intracellular calcium release, synaptic vesicle recycling, actin polymerization, cell proliferation, and insulin signaling.

### **CALCIUM SIGNALING: REGULATION OF Ins(1,4,5)P<sub>3</sub> AND Ins(1,3,4,5)P<sub>4</sub>**

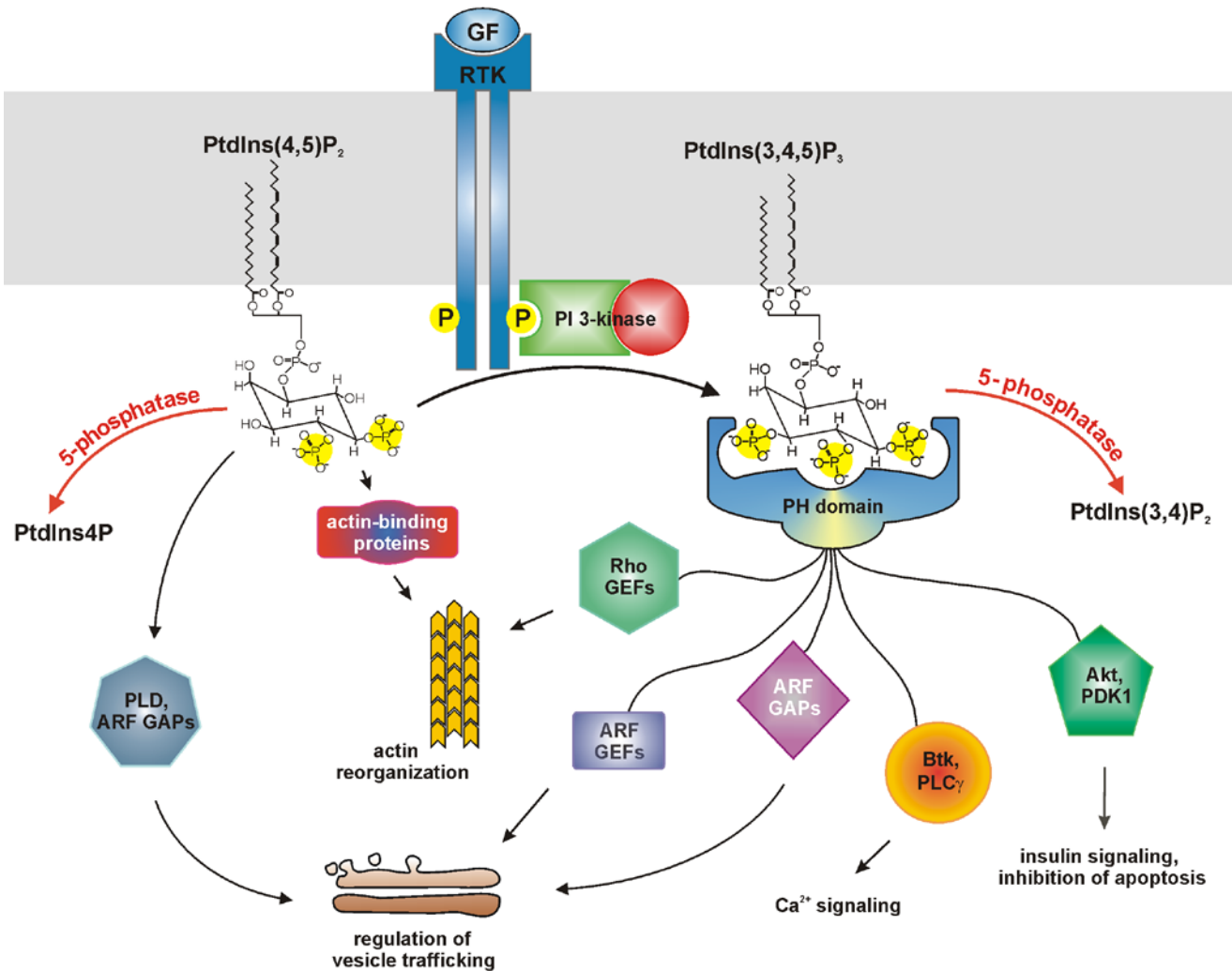
In nonexcitable cells Ins(1,4,5)P<sub>3</sub> stimulates calcium release from intracellular stores, followed by calcium influx across the

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<sup>1</sup>Abbreviations: PtdIns, phosphatidylinositol; PI 3-kinase, phosphoinositide 3-kinase; SHIP, SH2-containing inositol 5-phosphatase; OCRL, oculocerebrorenal syndrome of Lowe; SKIP, skeletal muscle and kidney-enriched inositol 5-phosphatase; PIPP, proline-rich inositol polyphosphate 5-phosphatase; BCR, B cell receptor; ITIM, immunoreceptor tyrosine-based inhibition motif.

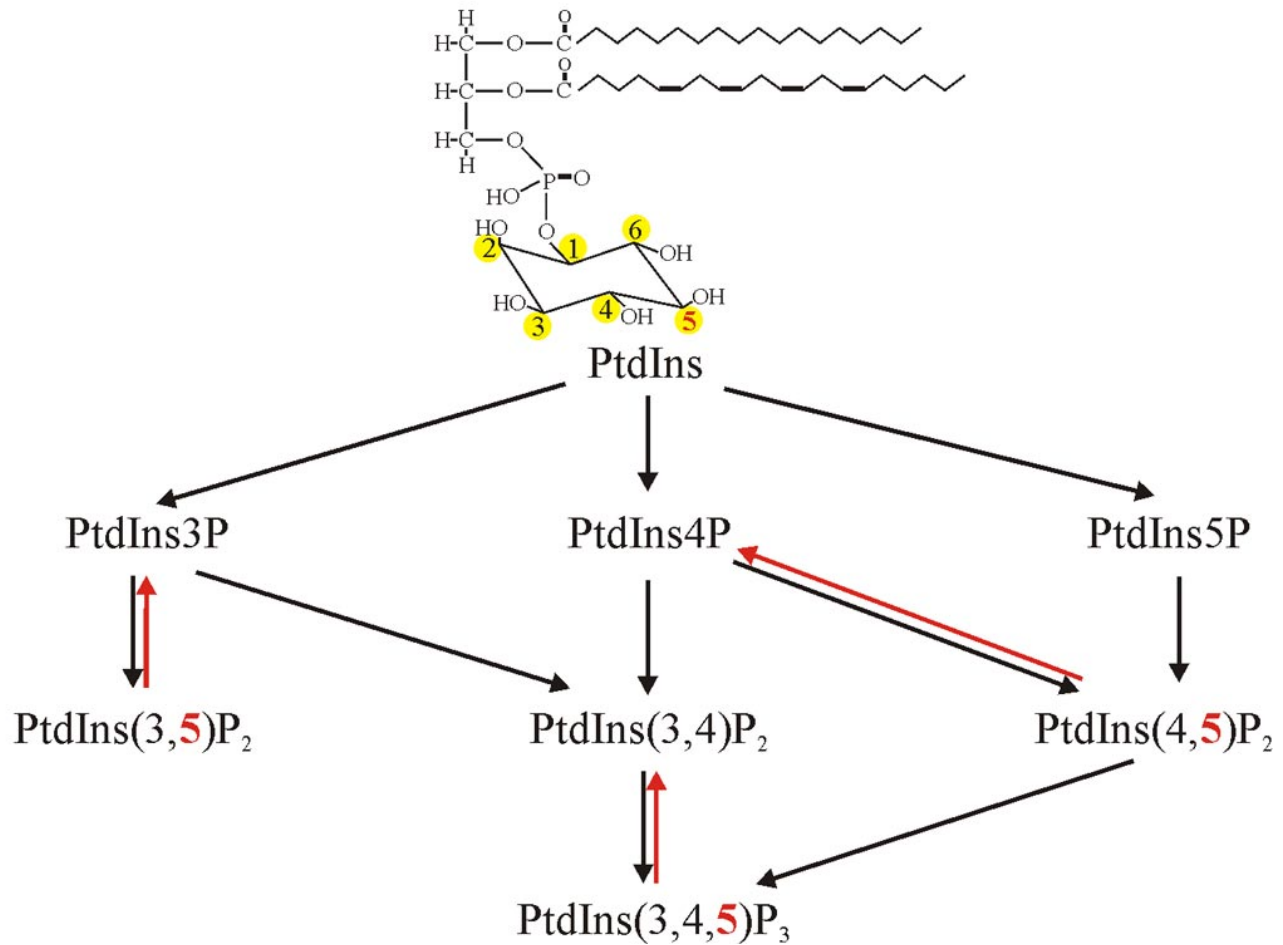


**Figure 1.** PtdIns(4,5)P<sub>2</sub> and PtdIns(3,4,5)P<sub>3</sub> signaling pathways. Growth factor (GF)-mediated activation of receptor tyrosine kinases (RTKs) leads to the activation of phosphoinositide 3-kinase (PI3-kinase) resulting in the synthesis of PtdIns(3,4,5)P<sub>3</sub> from PtdIns(4,5)P<sub>2</sub>. Both PtdIns(3,4,5)P<sub>3</sub> and PtdIns(4,5)P<sub>2</sub> are involved in many intracellular signaling pathways. PtdIns(4,5)P<sub>2</sub> interacts with a number of actin-binding proteins and promotes actin polymerization. It is also involved in the regulation of a number of proteins required for vesicle trafficking. PtdIns(3,4,5)P<sub>3</sub> interacts with and modulates the activities and localizations of a number of PH domain-containing proteins, which regulate anti-apoptotic, B-cell proliferation, insulin signaling, membrane trafficking, and actin reorganization.

plasma membrane via store operated channels (SOCs). Intracellular calcium regulates a wide range of cellular processes including gene transcription, contraction, **secretion**, and cell proliferation (10, 11). Ins(1,4,5)P<sub>3</sub> is rapidly metabolized, either by phosphorylation by a specific 3-kinase to Ins(1,3,4,5)P<sub>4</sub>, which facilitates calcium entry at the plasma membrane; or Ins(1,4,5)P<sub>3</sub> is dephosphorylated by the 5-phosphatases to form inositol 1,4-bisphosphate (Ins(1,4)P<sub>2</sub>), which does not release intracellular calcium (Fig. 3).

The 43-kDa 5-phosphatase (5-phosphatase I) is a widely expressed enzyme that hydrolyzes Ins(1,4,5)P<sub>3</sub> and Ins(1,3,4,5)P<sub>4</sub> forming Ins(1,4)P<sub>2</sub> and inositol 1,3,4-trisphosphate

(Ins(1,3,4)P<sub>3</sub>), respectively (12). This enzyme contains a central 5-phosphatase domain with a C-terminal CAAX motif that mediates localization of the enzyme to the inner wall of the plasma membrane, the site of Ins(1,4,5)P<sub>3</sub> production. Overexpression of the 43-kDa enzyme in CHO cells abrogates ATP-induced calcium oscillations (13). In contrast, underexpression of the 5-phosphatase using an anti-sense strategy results in increased Ins(1,4,5)P<sub>3</sub> levels, leading to increased sensitivity and amplitude of the calcium oscillations in response to both low- and high-dose agonist stimulation (14). Calcium oscillations regulate both activation of specific transcription factors and cell cycle progression. Cells underexpressing the 43-kDa 5-phosphatase



**Figure 2.** 5-Phosphatase metabolites. Phosphatidylinositol (PtdIns) consists of two acyl groups and an inositol ring attached to a glycerol backbone. The free hydroxyl groups of the inositol ring can be phosphorylated at positions 3, 4, and/or 5 by distinct kinases (black arrows) to generate various phosphoinositides. The action of 5-phosphatases (red arrows) modulates the signaling functions of these molecules. Inositol phosphates are not shown.

form tumors in nude mice (15). These studies demonstrate the critical role the 43-kDa 5-phosphatase plays in regulating intracellular calcium and thereby cell growth.

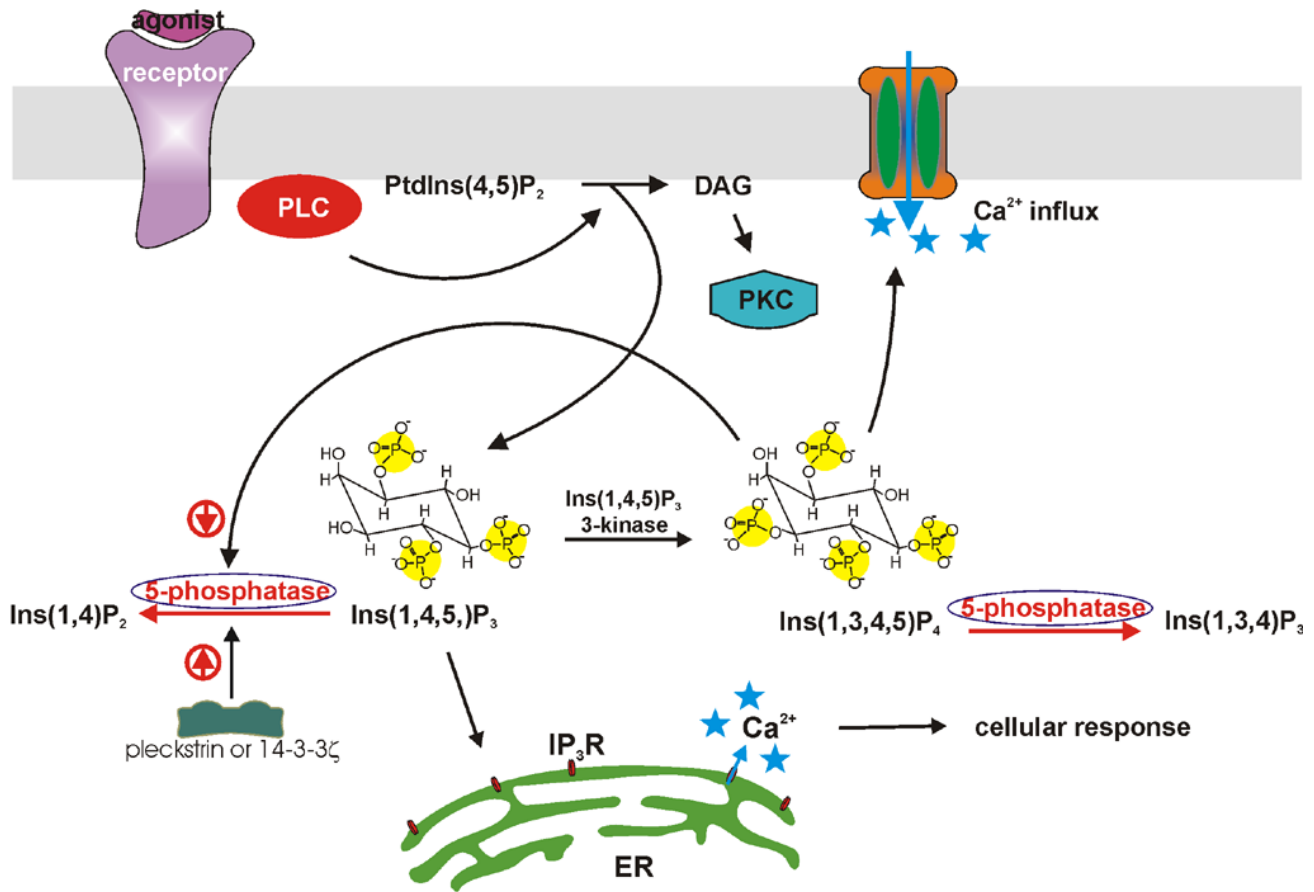
Recent exciting studies have shown Ins(1,3,4,5)P<sub>4</sub> inhibits the actions of the 43-kDa 5-phosphatase in hydrolyzing Ins(1,4,5)P<sub>3</sub> (16). The 43-kDa 5-phosphatase has a 10-fold higher affinity for Ins(1,3,4,5)P<sub>4</sub> than Ins(1,4,5)P<sub>3</sub> and a 100-fold lower  $V_{\max}$  for Ins(1,3,4,5)P<sub>4</sub>, compared to Ins(1,4,5)P<sub>3</sub>. Following agonist stimulation, Ins(1,3,4,5)P<sub>4</sub> has a much longer half-life than Ins(1,4,5)P<sub>3</sub>. Ins(1,3,4,5)P<sub>4</sub> inhibits Ins(1,4,5)P<sub>3</sub> metabolism by the 43-kDa 5-phosphatase and thereby facilitates calcium entry via activation of the store-operated Ca<sup>2+</sup> current (Fig. 3).

The 43-kDa 5-phosphatase forms a complex with the major platelet phosphoprotein pleckstrin following platelet activation. The association between these species results in enhanced 5-phosphatase activity (17). The 43-kDa 5-phosphatase also forms a complex in vitro, with the signaling adapter protein 14-3-3, resulting in increased Ins(1,4,5)P<sub>3</sub> 5-phosphatase

activity. The 43-kDa 5-phosphatase possesses a 14-3-3 binding site <sup>363</sup>RSESEE located at the C-terminus of the protein that mediates the interaction with 14-3-3 (18).

### Synaptic Vesicle Recycling

PtdIns(4,5)P<sub>2</sub> regulates the molecular events that govern endocytosis by mediating the recruitment of AP-2, AP180, dynamin, and epsin to the plasma membrane (7, 19), which leads to the formation of clathrin-coated pits and thereby initiates coated vesicle formation. The 5-phosphatase synaptojanin-1 plays a significant role in regulating synaptic vesicle endocytosis (20). Cortical neurons from synaptojanin-1-deficient mice demonstrate increased levels of PtdIns(4,5)P<sub>2</sub> and accumulate clathrin-coated vesicles. Electrophysiological studies of synaptojanin-1-deficient mice are consistent with impaired synaptic vesicle recycling (21). The *Caenorhabditis elegans* synaptojanin ortholog is encoded by the *UNC-26* gene. *Unc-26* mutants display defects in trafficking, vesicle budding, and uncoating, not



**Figure 3.** Role of the 43-kDa 5-phosphatase in regulating intracellular calcium release. The agonist stimulated hydrolysis of PtdIns(4,5)P<sub>2</sub> by phospholipase C (PLC) generates the second messengers Ins(1,4,5)P<sub>3</sub> and diacylglycerol (DAG). DAG activates protein kinase C (PKC), whereas Ins(1,4,5)P<sub>3</sub> binds Ins(1,4,5)P<sub>3</sub>-receptors (IP<sub>3</sub>R) on the endoplasmic reticulum (ER) and mobilizes Ca<sup>2+</sup>. Specific kinases can convert Ins(1,4,5)P<sub>3</sub> into Ins(1,3,4,5)P<sub>4</sub>, which is associated with extracellular Ca<sup>2+</sup> influx. Ins(1,3,4,5)P<sub>4</sub> inhibits the activity of the 43-kDa 5-phosphatase leading to sustained Ins(1,4,5)P<sub>3</sub> signal (32). Thus, 43-kDa 5-phosphatase enzyme activity may be increased via complex formation with either pleckstrin or 14-3-3. In contrast, Ins(1,3,4,5)P<sub>4</sub> inhibits the activity of the 43-kDa 5-phosphatase resulting in prolongation of the Ins(1,4,5)P<sub>3</sub> signal.

confined to the nervous system (22). Collectively, these studies suggest synaptojanin regulates the uncoating of clathrin-coated vesicles. The proposed mechanism for this effect is via metabolism of PtdIns(4,5)P<sub>2</sub>, which is essential for the recruitment of AP-2 and hence clathrin.

Two synaptojanin isoforms (1 and 2) have been cloned and characterized. Both have a similar domain structure containing an N-terminal Sac1 domain, a central 5-phosphatase domain and diverged C-terminal proline-rich domains. Synaptojanin-1 and -2 are expressed as 140- or 170-kDa isoforms and both isoforms demonstrate extensive splice variations in the C-terminal domains. Synaptojanin-1 is highly expressed in the brain, specifically in nerve termini; synaptojanin-2 is more widely expressed (23). Although in vitro synaptojanin-1 and -2, via the 5-phosphatase domain, hydrolyze PtdIns(4,5)P<sub>2</sub>, PtdIns(3,4,5)P<sub>3</sub>, Ins(1,4,5)P<sub>3</sub>, and Ins(1,3,4,5)P<sub>4</sub>, and the Sac1 domain serves as a polyphosphoinositide phosphatase, in vivo studies have

shown synaptojanin-1 regulates PtdIns(4,5)P<sub>2</sub> (24, 25). However, these studies must be interpreted with caution as there are significant difficulties in labeling phosphoinositides in cells, in particular PtdIns(3,4,5)P<sub>3</sub>, and secondly, PtdIns(4,5)P<sub>2</sub> is generated in vast excess of PtdIns(3,4,5)P<sub>3</sub>. It is of interest to note that in vitro phosphatase assays have demonstrated pronounced differences in PtdIns(3,4,5)P<sub>3</sub> 5-phosphatase activity in synaptojanin-deficient mice compared to normal litter mate controls (21).

Synaptojanin-1 hydrolyzes PtdIns(4,5)P<sub>2</sub> bound to the actin regulatory proteins α-actinin, vinculin, gelsolin, and profilin, and decreases the number of stress fibers in the cell (26). Synaptojanin-1 has been shown in vitro to interact with the SH3 domain of the dynamin-binding protein: synapin I. The latter also binds to N-WASP (neuronal Wiskott-Aldrich syndrome protein), a PtdIns(4,5)P<sub>2</sub>-binding protein enriched in nerve terminals, which promotes actin polymerization (27).

The synaptojanin-1 C-terminal proline-rich domain mediates interactions with numerous SH3 domain-containing proteins including amphiphysin, and the low molecular weight small SH3 domain-containing proteins SH3p4, SH3p8, and SH3p13, also known as endophilins (28, 29). Binding of synaptojanin with the nerve terminal proteins intersectin and synapin has been shown *in vitro* but not *in vivo* (27, 30). The 170-kDa synaptojanin isoform also complexes with the clathrin adapter AP-2 and with the Epsin15 homology (EH) domain of the clathrin-associated protein Eps15 (31, 32).

Synaptojanin-1, dynamin, amphiphysin, epsin, Eps15, and AP180 are phosphorylated in resting nerve terminals and are coordinately dephosphorylated upon endocytosis of synaptic vesicles. Phosphorylation may regulate protein-protein interactions during synaptic vesicle recycling (33, 34).

Synaptojanin-2 may also play a role in regulating endocytosis. The activated small GTPase Rac1 binds synaptojanin-2. Synaptojanin-2 directly interacts with Rac1 in a GTP-dependent manner, resulting in translocation of the 5-phosphatase to membrane ruffles and inhibition of endocytosis (35). Catalytically active synaptojanin-2, which is localized to the plasma membrane, inhibits endocytosis of the EGF receptor, suggesting synaptojanin-2 mediates Rac1 inhibition of endocytosis. However, membrane-targeted synaptojanin-2 had no effect on lamellipodia, which is regulated by PI 3-kinase and Rac1 activity (35). One of the splice variants of synaptojanin-2 is localized to the outer mitochondrial membrane, whereby the 5-phosphatase forms a complex with a novel integral mitochondrial membrane protein designated OMP25. This interaction regulates the intracellular distribution of the organelle (36).

### **Regulation of Hemopoietic Cell Activation and Proliferation**

The SH2 domain-containing 5-phosphatase-1 (SHIP-1) suppresses PI 3-kinase-dependent pathways following recruitment to immunoreceptor tyrosine-based inhibition motif (ITIM)-containing proteins (Fig. 4). Targeted disruption of murine SHIP-1 results in a myeloproliferative-like syndrome characterized by a massive increase in granulocyte macrophage progenitor cells in the bone marrow and spleen, leading to dramatic myeloid infiltration of the lungs, wasting and shortened life span (37, 38). Progenitor cells from SHIP-1<sup>-/-</sup> mice demonstrate hyperresponsiveness to many hematopoietic growth factors and cytokines, compared to SHIP-1<sup>+/+</sup> littermates (37). Collectively, these findings are consistent with a model in which SHIP-1 acts as a negative regulator of hematopoietic cell proliferation and survival.

SHIP-1 hydrolyzes PtdIns(3,4,5)P<sub>3</sub> and Ins(1,3,4,5)P<sub>4</sub>, forming PtdIns(3,4)P<sub>2</sub> and Ins(1,3,4)P<sub>3</sub>, respectively. The C-terminal proline-rich domain is essential for full enzyme activity, presumably by regulating membrane association (39, 40). More recent studies have demonstrated SHIP-1 also hydrolyzes PtdIns(4,5)-P<sub>2</sub>, forming PtdIns 4-P. However, its principal *in vivo* phosphoinositide substrate is likely to be PtdIns(3,4,5)P<sub>3</sub> (41). SHIP-1 is

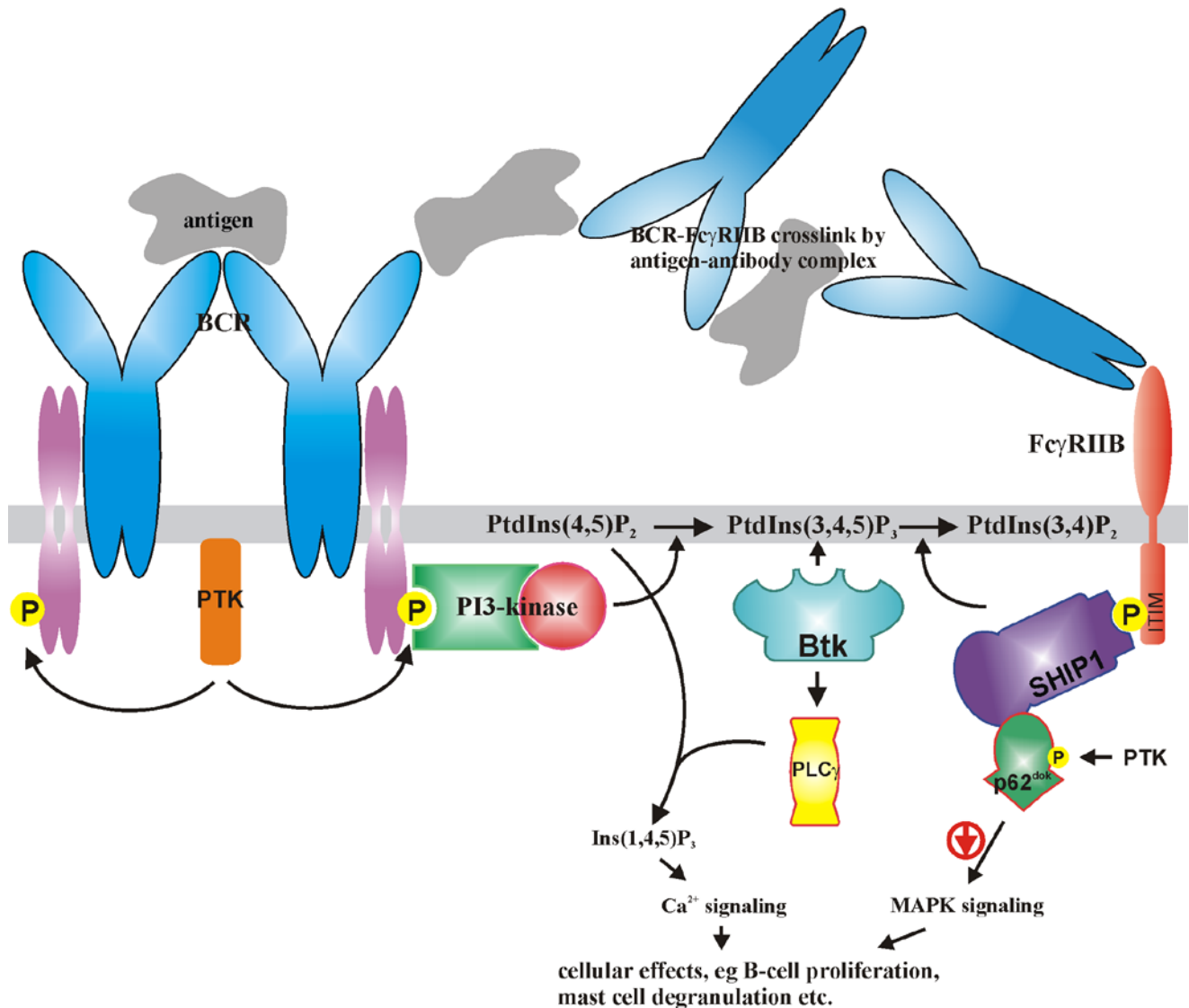
composed of multiple domains including an N-terminal SH2 domain, central catalytic 5-phosphatase domain, two NPXY motifs that mediate binding to phosphotyrosine binding (PTB) domain-containing proteins, and proline-rich regions in the C-terminus (42, 43). SHIP-1 is expressed exclusively in cells of hematopoietic origin and has been shown to play a significant functional role in regulating B cell, mast cell, and phagocytic cell activation (44).

Engagement of antigen receptors such as the B cell receptor (BCR), T cell receptor, FcγRIII, and FcεRI results in synthesis of PtdIns(3,4,5)P<sub>3</sub>, which activates Bruton's tyrosine kinase and phospholipase C leading to intracellular calcium release. In B cells, coligation of the B cell antigen receptor with FcγRIIB initiates phosphorylation of the FcγRIIB ITIM, which results in the recruitment of SHIP-1 leading to PtdIns(3,4,5)P<sub>3</sub> hydrolysis and attenuation of downstream signaling (56). SHIP-1 also recruits the RasGAP-binding protein p62<sup>dok</sup> to the FcγRIIB-BCR-complex, resulting in its phosphorylation and increased binding to RasGAP. This association is critical for inhibition of BCR-mediated activation of the Ras signaling pathway (45) (Fig. 4).

In mast cells, SHIP-1 plays a significant role in regulating mast cell degranulation by limiting IgE-induced degranulation and preventing Steel Factor-induced intracellular signaling from initiating degranulation (46). In mast cells, SHIP-1 is recruited to the ITIM of the FcεRI/ FcγRIIB complex (47). The mechanism of SHIP-1 action appears to be the same in both mast and B cells and functions to regulate cytokine- or antigen-induced stimulation of PtdIns(3,4,5)P<sub>3</sub> synthesis and thereby extracellular calcium entry and activation of the survival factor, serine-threonine kinase Akt. The SHIP-1 C-terminal proline-rich domain plays an important functional role in mediating these responses, as intact SHIP-1 but not SHIP-1 truncation mutants that lack the C-terminal proline-rich domain, rescue SHIP-1<sup>-/-</sup> bone marrow-derived mast cell hyperresponsive phenotype (40).

In phagocytic cells such as macrophages, SHIP-1 regulates phagocytosis. Macrophages overexpressing SHIP-1 show impaired CR3-mediated phagocytosis, and cells from SHIP-1<sup>-/-</sup> mice demonstrated 2.5-fold enhanced phagocytosis (48). SHIP-1 is expressed as 145-, 135-, 125-, and 110-kDa isoforms resulting from C-terminal truncation, with the expression of the different isoforms correlating with the stage of hematopoietic differentiation and leukemogenesis (49, 50). In addition, the cytokine hyperresponsive phenotype of the SHIP-1<sup>-/-</sup> mice is similar to that detected following Bcr-Abl transformation in mice (51, 52). The Bcr-Abl oncogene causes a clonal expansion of hematopoietic progenitor cells and granulocyte lineage cells, resulting in chronic myeloid leukemia (CML). An inverse correlation between the expression of the oncogenic fusion protein Bcr-Abl and SHIP-1 in chronic myeloid leukemia has been detected, suggesting decreased SHIP-1 activity may contribute to the proliferative advantage of these leukemic cells (53). SHIP-1 expression may be regulated by the Bcr-Abl oncogenic fusion protein, as treatment of CML cells with the specific tyrosine kinase inhibitor STI571 results in re-expression of SHIP-1. It





**Figure 4.** SHIP-1 regulation of B-cell negative signalling. Activation of PI3-kinase triggered by B-cell receptor (BCR) crosslinking with antigens leads to PtdIns(3,4,5)P<sub>3</sub> synthesis. Membrane recruitment of Bruton's tyrosine kinase (Btk) via its PH domain results in activation of phospholipase C (PLC), subsequent PtdIns(4,5)P<sub>2</sub> hydrolysis and intracellular Ca<sup>2+</sup> release, and B-cell proliferation. Co-engagement of FcγRIIB with BCR by specific antibodies leads to recruitment of SHIP-1 to the phosphorylated ITIM of FcγRIIB, PtdIns(3,4,5)P<sub>3</sub> hydrolysis, and attenuation of the proliferative signaling. The recruitment of p62<sup>dok</sup> to ITIM-bound SHIP-1 also leads to phosphorylation by tyrosine kinases (PTK) of p62<sup>dok</sup> and down-regulation of MAPK signaling.

is noteworthy that the more widely expressed and homologous 5-phosphatase SHIP-2 interacts with the Bcr-Abl oncogene product. In addition, SHIP-2 is constitutively tyrosine phosphorylated and associated with Shc in chronic myeloid leukemic progenitor cells, suggesting a role for SHIP-2 in 210<sup>bcr/abl</sup>-mediated myeloid expansion (54). Both SHIP-1 and SHIP-2 are constitutively tyrosine phosphorylated in Bcr-Abl-expressing cells. SHIP-2 specifically binds the SH3 domain of Abl, but not Grb2, whereas SHIP-1 binds the SH3 domains of Src and Grb2.

Finally, an intriguing study has demonstrated the gene *DSHP* (Duncan's disease SH2 domain protein) is responsible for X-linked lymphoproliferative syndrome, which is characterized by increased susceptibility to Epstein-Barr virus infection, leading to high-grade lymphoma. *DSHP* comprises a single SH2 domain that demonstrates extensive sequence homology with the SH2 domain of SHIP-1. The results of this study suggest that *DSHP* may compete with SHIP-1 binding to sites of activated receptors and thereby enhance lymphocyte activation and proliferation (55).

### Regulation of Insulin Signaling

SHIP-2 is a widely expressed 5-phosphatase that plays a significant role in negatively regulating insulin signaling (56, 57). SHIP-2 was originally cloned and named “inositol polyphosphate 5-phosphatase-like protein 1” (INPPL1) in the search for functional complementation of Fanconi’s anemia (58). Subsequent studies have suggested the diversity within the *N*-terminal SH2 domains of SHIP-2 and INPPL1 probably relate to a cloning artefact (59). SHIP-2 contains an amino-terminal SH2 domain, a central 5-phosphatase domain, and a carboxyl-terminal proline-rich domain and bears significant sequence identity with the 5-phosphatase, SHIP-1, except in the proline-rich domain. SHIP-2 C-terminal domain predicts for a PDZ-binding domain as well as sites for tyrosine phosphorylation and a SAM domain (sterile-alpha module). Unlike SHIP-1, SHIP-2 is widely expressed in many tissues and cells. SHIP-2 hydrolyzes the 5-position phosphate from  $\text{PtdIns}(3,4,5)\text{P}_3$  and  $\text{PtdIns}(4,5)\text{P}_2$  and in some, but not all studies, has been shown to hydrolyze the soluble inositol phosphate  $\text{Ins}(1,3,4,5)\text{P}_4$  (54, 60, 61). SHIP-2 undergoes cytokine-, growth factor-, and insulin-stimulated phosphorylation in a number of cell lines (54, 62). SHIP-2 localizes to focal contacts of adherent cells, as well as lamellipodia of EGF-stimulated cells. Membrane association is in part mediated via SHIP-2 association with the  $\text{p130}^{\text{Cas}}$  adapter protein at focal adhesions, regulating cell spreading. Association between these species is dependent on the SHIP-2 SH2 domain and is enhanced by tyrosine phosphorylation and cell adhesion (63). In addition SHIP-2, like PTEN, regulates both  $\text{PtdIns}(3,4,5)\text{P}_3$ -mediated Akt activation and induces cell cycle arrest, associated with increased stability of expression of the cell cycle inhibitor  $\text{p27}^{\text{KIP1}}$  (61).

Recent studies have demonstrated SHIP-2 negatively regulates insulin signaling. Insulin-activated Akt activation and translocation of the glucose transporter GLUT4 in both skeletal muscle and adipocytes are largely dependent on PI 3-kinase activation and subsequent production of  $\text{PtdIns}(3,4,5)\text{P}_3$  and  $\text{PtdIns}(3,4)\text{P}_2$  (64, 65). Microinjection of recombinant SHIP-1 into 3T3-L1 adipocytes inhibited insulin-induced GLUT4 translocation, DNA synthesis, and membrane ruffling responses and in addition inhibited downstream signaling pathways from the insulin receptor (66). Because SHIP-1, in contrast to SHIP-2, is not expressed in either skeletal muscle or fat, subsequent studies have concentrated on investigating the role SHIP-2 plays in regulating insulin signaling. Both endogenous and recombinant SHIP-2 are tyrosine phosphorylated in response to insulin stimulation, associated with a reduction in mitogen-activated protein kinase (MAPK) activation (56). Overexpression of wild-type or dominant-negative SHIP-2 in fibroblasts, or 3T3-L1 adipocytes, does not affect early signaling events following insulin stimulation such as insulin receptor substrate (IRS) phosphorylation, or PI 3-kinase activation. However,  $\text{PtdIns}(3,4,5)\text{P}_3$  levels are significantly increased in cells expressing dominant-negative SHIP-2 compared to cells expressing the wild-type enzyme (67). In addition, overexpression of SHIP-2, but not catalytically inactive

SHIP-2, in 3T3-L1 adipocytes results in negative regulation of insulin-induced signaling. Cells expressing dominant-negative SHIP-2 are unable to down-regulate responses downstream of PI 3-kinase including Akt and  $\text{PKC}\lambda$  activation, GLUT4 translocation, glucose uptake, and glycogen synthase kinase  $3\beta$  phosphorylation (67).

Newborn homozygous mice lacking SHIP-2 exhibit severe hypoglycemia, show aberrant expression of genes involved in gluconeogenesis in the liver, and die in the neonatal period from respiratory failure (57). SHIP-2<sup>+/-</sup> and <sup>+/+</sup> mice exhibit similar blood glucose levels under both feeding and fasting conditions and produce comparable levels of insulin in response to glucose. However, adult SHIP-2 heterozygous mutant mice demonstrate insulin sensitivity associated with increased translocation of GLUT4 to the plasma membrane, in response to insulin treatment (57). Although the signaling pathways mediating the phenotype of insulin-hypersensitivity have yet to be fully determined, insulin-stimulated GLUT4 translocation to the plasma membrane appears to be enhanced in mice lacking SHIP-2.

### Ocular Cerebrorenal Syndrome of Lowe (OCRL)

Lowe’s syndrome is an X-chromosome-linked disorder characterized by a pleiotropic phenotype that includes mental retardation, congenital cataracts and renal tubular acidosis (68). The gene locus disrupted in this syndrome was mapped to Xq25-q26 and subsequently found to encode a 5-phosphatase with a predicted molecular weight of 112 kDa. The OCRL (oculocerebrorenal syndrome of Lowe) gene product is most closely related to the 75-kDa 5-phosphatase (5-phosphatase II), sharing 71% amino acid sequence similarity (69). The recombinant enzyme preferentially hydrolyzes  $\text{PtdIns}(4,5)\text{P}_2$ , but activity towards  $\text{PtdIns}(3,4,5)\text{P}_3$ ,  $\text{Ins}(1,4,5)\text{P}_3$ , and  $\text{Ins}(1,3,4,5)\text{P}_4$  has been demonstrated (70, 71). Cell lines from patients with Lowe’s syndrome accumulate  $\text{PtdIns}(4,5)\text{P}_2$ , consistent with the enzyme acting predominantly as a  $\text{PtdIns}(4,5)\text{P}_2$  5-phosphatase (71). However, it is noteworthy that although other 5-phosphatases are expressed in these cells, they appear unable to compensate for the loss of the OCRL 5-phosphatase. Surprisingly, mice that lack Lowe’s protein demonstrate no phenotype (72). The localization of the OCRL protein has been proposed to be in either the Golgi (human fibroblasts), or the lysosomes (human kidney proximal tubules) (71, 73). The lysosomal localization of the protein is supported by the intriguing observation that several lysosomal enzymes are increased in the plasma of OCRL patients, presumably as the result of abnormal lysosomal trafficking (74).

Lowe’s syndrome is the first and only known inborn error of phosphatidylinositol metabolism. The relationship between the defective enzyme and the clinical phenotype of the disease is not well understood. Mutations in the OCRL gene in these patients are heterogeneous. Deletions, frame shifts, nonsense, and missense mutations throughout the coding sequence, including outside the 5-phosphatase catalytic domain have been reported (75–78).

### 5-Phosphatases of Unknown Function

Recent studies in the last 2 years have resulted in the cloning of three new mammalian 5-phosphatases, a 52-kDa enzyme designated "SKIP" (for skeletal muscle and kidney enriched 5-phosphatase), a 72-kDa protein called "pharbin," and 107-kDa 5-phosphatase called "PIPP" (for proline-rich inositol polyphosphate 5-phosphatase). The latter 5-phosphatase has extensive proline-rich regions within both the *N*- and *C*-terminal domains. Within these regions, six putative 14-3-3 binding sites have been identified, suggesting this isoform, like the 43-kDa 5-phosphatase, complexes with this signaling adapter protein. PIPP is widely expressed and recombinant PIPP demonstrates activity towards  $\text{Ins}(1,4,5)\text{P}_3$ ,  $\text{Ins}(1,3,4,5)\text{P}_4$ , and  $\text{PtdIns}(4,5)\text{P}_2$ ; however, metabolism of  $\text{PtdIns}(3,4,5)\text{P}_3$  and  $\text{PtdIns}(3,5)\text{P}_2$  have not been reported (79). The 107-kDa enzyme may undergo posttranslational modification such as serine/threonine phosphorylation. This enzyme localizes at membrane ruffles, which requires the presence of both the *N*- and *C*-terminal proline-rich domains. However, this enzyme localization does not affect membrane ruffling.

SKIP is expressed as three splice variants, the largest of which is 52 kDa. Although the 5-phosphatase is highly expressed in both kidney and skeletal muscle, the enzyme appears to be widely expressed at similar levels in many tissues. Recombinant SKIP hydrolyzes both soluble inositol phosphates  $\text{Ins}(1,4,5)\text{P}_3$  and  $\text{Ins}(1,3,4,5)\text{P}_4$ , and phosphoinositides  $\text{PtdIns}(4,5)\text{P}_2$  and  $\text{PtdIns}(3,4,5)\text{P}_3$ . Recombinant and endogenous SKIP localize in a cytosolic distribution and concentrate in a perinuclear area. Overexpression of SKIP results in reduced actin stress fiber formation in these areas (80).

Cloning and characterization of human, mouse, and rat isoforms of a 72-kDa 5-phosphatase designated "pharbin" for phosphatase that induces arborization (81) have been reported by three groups. The human isoform has been designated the "Type IV 5-phosphatase," due to its unique substrate specificity (41) and we have designated the mouse isoform the 72-kDa 5-phosphatase (82). The pharbin sequence bears 88% amino acid sequence identity with the mouse 72-kDa 5-phosphatase, whereas the human Type IV enzyme recently identified by Kisseleva et al. (41) demonstrates 76% amino acid identity over 649 amino acids with the mouse enzyme, suggesting it may represent either the human homolog, or a spliced variant. Considerable sequence diversity occurs within the *N*-terminal proline-rich domain, with the human isoform containing a greater content of prolines (19% compared to 13% in the mouse ortholog). Amino acid sequence analysis has demonstrated that in addition to an *N*-terminal proline-rich domain, the sequence predicts for a *C*-terminal CAAX motif, and a putative ITAM (immunoreceptor tyrosine-based activation motif). The latter is commonly found in the cytoplasmic region of many immune receptors and functions as a link between activated cell surface receptors and their effector molecules in the cytosol. Whether this motif plays a functional role for this 5-phosphatase has yet to be shown. The mouse 72-kDa enzyme is highly expressed in

brain and testis, with the human enzyme demonstrating in addition high expression in heart, with lower levels in liver and kidney (41, 82). The 72-kDa 5-phosphatase localizes to the cytoplasmic surface of the Golgi, mediated by the *N*-terminal proline-rich domain (82). In contrast, pharbin localizes to the plasma membrane of neural cells and promotes arborization (81). Different substrate specificities have been reported for the mouse, rat, and human 72-kDa 5-phosphatases. Studies using the recombinant rat isoform suggest the enzyme hydrolyzes  $\text{Ins}(1,4,5)\text{P}_3$ ,  $\text{Ins}(1,3,4,5)\text{P}_4$ , and  $\text{PtdIns}(4,5)\text{P}_2$ , yet in contrast neither the mouse nor human 5-phosphatase were shown to hydrolyze the soluble inositol phosphates (81), and demonstrate greatest activity towards  $\text{PtdIns}(3,4,5)\text{P}_3$  (41, 82). In a detailed kinetic analysis, Majerus and colleagues have shown the human type IV enzyme has the greatest activity towards  $\text{PtdIns}(3,4,5)\text{P}_3$  of any mammalian 5-phosphatase including SHIP-1, with a  $K_m$  of 0.65 M versus 5.95 M, respectively. In these studies, both SHIP-1 and 72-kDa 5-phosphatase enzyme activity was shown to be highly sensitive to the presence of specific detergents in the assay, which may in part explain some discrepancies in the literature about the distinct substrate specificity of various 5-phosphatase isoforms (41).

### Insights From the Analysis of Yeast 5-Phosphatases

The yeast *Saccharomyces cerevisiae* has been utilized by several investigators as a model organism to study 5-phosphatase cellular function. Four 5-phosphatases have been identified and partially characterized in yeast. Three enzymes designated Inp51p, Inp52p, and Inp53p contain an *N*-terminal Sac1 domain, a central 5-phosphatase domain, and a *C*-terminal proline-rich domain. Inp54p contains a 5-phosphatase domain and a *C*-terminal hydrophobic tail (83). All of these 5-phosphatases hydrolyze  $\text{PtdIns}(4,5)\text{P}_2$  (84–87). Inp52p also demonstrates 5-phosphatase activity towards  $\text{PtdIns}(3,5)\text{P}_2$  (87). Null mutation of any one of the Sac1 domain-containing 5-phosphatases results in little phenotype (88), suggesting that these 5-phosphatases have overlapping functions. However, disruption of any two of the Sac1 domain-containing 5-phosphatases results in marked phenotypic changes characterized by cell wall thickening and actin depolymerization defects, yet deletion of all three is lethal (84, 88, 89).

In yeast, the actin cytoskeleton appears to be linked to endocytosis [reviewed by (90)] and null mutation of *inp51 inp52* results in defects of the actin cytoskeleton as well as severe defects in fluid-phase and receptor-mediated endocytosis (84, 89). Deletion of any two yeast Sac1 domain-containing 5-phosphatases results in a phenotype that includes disorganization of polymerized actin and delocalization of actin patches from the growing yeast bud to the daughter cell [reviewed by Hughes et al. (91)]. It is noteworthy the yeast 5-phosphatases, Inp52p and Inp53p, translocate to actin patches upon osmotic stress, the site of plasma membrane invaginations. In addition, overexpression of Inp52p and Inp53p, but not catalytically inactive Inp52p, results in a significant reduction in the repolarization



time of actin patches following osmotic stress (87). Deletion of *inp51 inp52* results in massive plasma membrane invaginations that may reflect an imbalance between endocytosis and exocytosis (89). However, exocytosis appears normal in all Sac1 5-phosphatase mutant strains, as secretion of invertase or a reporter protein is not impaired (83, 89). The *inp51 inp53* and *inp52 inp53* null mutants also display impairment of endocytosis and plasma membrane invaginations; however, the phenotype is less marked than that observed in the *inp51 inp52* null mutant strain (88, 89).

Defects in endocytosis and the actin cytoskeleton appear to correlate with abnormalities in cell polarity in the *inp51 inp52* and *inp52 inp53* disrupted strains, which demonstrate abnormal bud site selection (89). All three double null mutants also show thickened cell walls, compromised growth on hyperosmotic media, and fragmented vacuoles (84, 87, 88).

Yeast 5-phosphatases have been shown to display genetic interactions with several other proteins. Deletion of *inp51* is capable of partially suppressing the growth defect of a *sac6* null mutant. *SAC6* encodes the yeast homolog of fimbrin, an actin bundling protein, suggesting that Inp51p and Sac6p have opposing effects on the actin cytoskeleton (89). Double deletion of *inp51* together with *pan1*, the yeast homolog of eps15, is lethal (92). Pan1p is required for both endocytosis and organization of the actin cytoskeleton (93, 94).

Genetic screening for suppressors of *pma1* that divert mutant Pma1p from delivery to the vacuole have identified Inp53p (Sop2) (95). Results from this study have implicated Inp53p in the regulation of trafficking between the endosome and Golgi. Inp53p has also been predicted to play a role in clathrin-coated vesicle trafficking from the *trans* Golgi network, based on the demonstration of the 5-phosphatase's genetic interaction with the clathrin heavy chain gene *CHC1* (96).

Although the Sac1 domain-containing 5-phosphatases appear to play a significant role in regulating organization of the actin cytoskeleton and thereby endocytosis, the yeast 5-phosphatase Inp54p, is localized to the endoplasmic reticulum and regulates secretion from this compartment. Inp54p localizes to the endoplasmic reticulum by its C-terminal hydrophobic tail, which is consistent with a tail-anchored protein. Deletion of *inp54* leads to a 2- to 3-fold increase in secretion of a reporter protein, compared to wild-type cells or *inp51*, *inp52*, or *inp53* null mutants (83). Consistent with its localization to the endoplasmic reticulum, loss of *inp54* does not affect the actin cytoskeleton, cell polarity, or the morphology of the cell (83).

To date, few proteins that physically interact in vivo with the yeast 5-phosphatases have been identified. High-throughput yeast two-hybrid screens have identified Gtt1p and the uncharacterized Ydl204p as potential binding partners for Inp54p (97). Gtt1p is a glutathione S-transferase protein localized to the endoplasmic reticulum (98) placing it in proximity to Inp54p. Glutathione S-transferases are thought to play a role in cellular detoxification, although Gtt1p is not required for protection against oxidative stress (98).

## CONCLUSIONS

The 5-phosphatases demonstrate a regulatory function in many cellular processes, and although the potential role these enzymes play in human diseases is only recently emerging, results of many recent studies have demonstrated the critical role 5-phosphatase-mediated hydrolysis of both inositol phosphate and phosphoinositide signaling molecules plays in the regulation of cellular activation and proliferation.

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